

unmodified server (Redis · Memcached · Nginx · Node · PostgreSQL)

POSIX syscalls: sockets · time · randomness · threads



OneBarrier determinism libOS

LD_PRELOAD · no kernel · no app change

virtual clock
(time)

rng trap
(getrand/urnd)

share-nothing
shard

replay + output
suppress

durable ordered log + timestamp- T snapshot



**1Pipe fabric:
in-network total order +
reliable 2PC commit barrier**

decentralized recovery cut



**RDMA replica
(durability)**

durability = 1-RTT replication